

Choose efficient representations

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity. Copy or complete the first step.

2. Three fifths of a number is 84. Find 35% of the number.

3. A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?

4. Miro says: Miro starts calculating before deciding what the problem asks and which information matters. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity. Copy or complete the first step.
Answer 244.8.
2. Three fifths of a number is 84. Find 35% of the number.
Answer 49.
3. A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?
Answer 320 cm³.
4. Miro says: Miro starts calculating before deciding what the problem asks and which information matters. Is this correct? Explain.
Answer Represent the relationships first, label each subtotal and test the final answer against every condition.

Choose efficient representations

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

6. Check this result using an inverse, estimate or known fact: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?

2. Three fifths of a number is 84. Find 35% of the number.

7. Prove or explain your reasoning: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

3. A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?

8. Identify and correct this misconception: Miro starts calculating before deciding what the problem asks and which information matters.

4. Solve this independently and show every stage: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Three fifths of a number is 84. Find 35% of the number.

10. Write and solve a similar question to: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?

Teacher answers and checking prompts

1. A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.
Answer 244.8.
2. Three fifths of a number is 84. Find 35% of the number.
Answer 49.
3. A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?
Answer 320 cm³.
4. Solve this independently and show every stage: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.
Answer 244.8.
5. Use a second method or representation for: Three fifths of a number is 84. Find 35% of the number.
Answer Expected result: 49. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?
Answer Expected result: 320 cm³. A valid check must be shown.
7. Prove or explain your reasoning: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?
Answer 120 pupils. The explanation must show why the method works.
8. Identify and correct this misconception: Miro starts calculating before deciding what the problem asks and which information matters.
Answer Represent the relationships first, label each subtotal and test the final answer against every condition.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Choose efficient representations

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?

6. Create a new example that tests the same idea as: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?

2. Prove the answer to this question, rather than only calculating it: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Three fifths of a number is 84. Find 35% of the number.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 320 cm³.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro starts calculating before deciding what the problem asks and which information matters.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A pie-chart sector is 126 degrees and represents 42 pupils. How many pupils are in the survey?
Answer 120 pupils. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A quantity increases from 240 by 15%, then decreases by 20%. Find the final quantity.
Answer Expected result: 244.8. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Three fifths of a number is 84. Find 35% of the number.
Answer Expected result: 49. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 320 cm³.
Answer One valid reconstruction is: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro starts calculating before deciding what the problem asks and which information matters.
Answer Represent the relationships first, label each subtotal and test the final answer against every condition.
6. Create a new example that tests the same idea as: A cuboid is 12 cm by 8 cm by 5 cm. A third of its volume is filled. What volume remains empty?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.